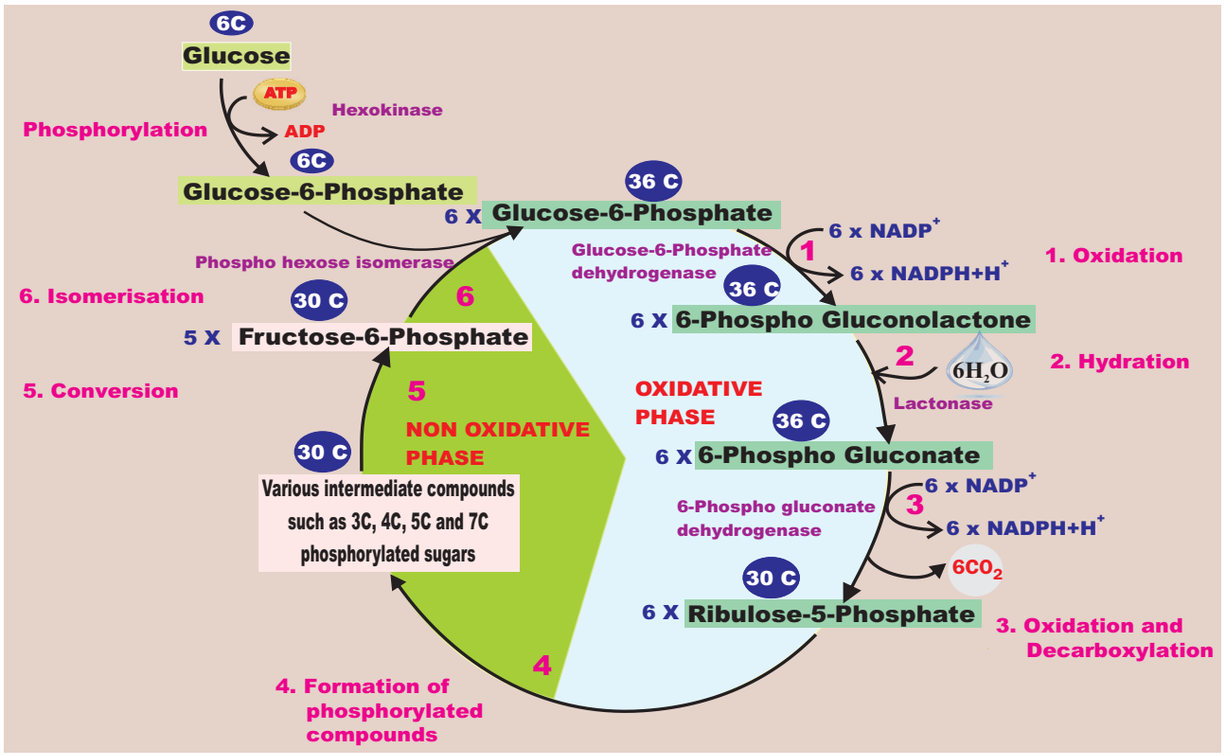


Figure 14.16: Pentose phosphate pathway or HMP shunt

phosphate pathway is controlled by glucose-6-phosphate dehydrogenase enzyme which is inhibited by high ratio of NADPH to $NADP^+$.

Significance of pentose phosphate pathway

1. HMP shunt is associated with the generation of two important products, NADPH and pentose sugars, which play a vital role in anabolic reactions.
2. Coenzyme NADPH generated is used for reductive biosynthesis and counter damaging the effects of oxygen free radicals
3. Ribose-5-phosphate and its derivatives are used in the synthesis of DNA, RNA, ATP, NAD^+, FAD and Coenzyme A.
4. Erythrose is used for synthesis of anthocyanin, lignin and other aromatic compounds.
5. It plays a role on fixation of Co_2 in photosynthesis through RUBP

Summary

Respiration is a biological process in which energy is released by breaking down of complex organic substances into simple compounds. The respiratory substrates may be carbohydrate, protein or fats. Respiration is of two types, aerobic (with O_2) and anaerobic (without O_2). All plants, animals and most of the microbes derive energy from aerobic respiration. Some bacteria and fungi like yeast show anaerobic respiration. Aerobic respiration consists of four stages and they are glycolysis, link reaction, TCA cycle and ETS. Glycolysis is the first stage which occurs in cytosol and common for both aerobic and anaerobic respiration and it involves breaking down of glucose into two molecules of pyruvic acid. Acetyl CoA formed from pyruvic acid, acts as a link between glycolysis and Krebs cycle. Krebs cycle takes place in matrix of mitochondria and also called as citric acid cycle in which CO_2 and H_2O were produced. Hydrogen removed from the substrates is received by coenzymes which get reduced. They are again oxidised by removal of hydrogen. This hydrogen splits into protons and electrons. The electrons transferred through various electron transport carriers present in inner membrane of mitochondria is used for the synthesis of ATP with the help of ATP synthase. This process is called **oxidative phosphorylation**.

Anaerobic respiration involves incomplete breaking down of the substrate glucose into ethyl alcohol or lactic acid. In aerobic respiration 36 ATP molecules are produced in plant

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